Lauds Factories - Co-funded by the European Union, 2024-2026 GA 101135986

LAUDS FACTORIES

Open Call 2 | LAUDS Replication

Collaboration Report

DATA-RIDE

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Authors

Artur Kim Shum
Laura Sobral
Ricardo Ruiz

Project coding LAUDSREP-01-19	*DATA-RIDE: Data and Advocacy for Transforming Active Riding & Infrastructure for a Democratic Environment*	
Deliverable title Production Process Blueprint: Data-Ride		**Due date** 01.10.2026
Hybrid Team entities **Lead partner:** re:edu GmbH & Co. KG **Partner 1: Laura Sobral** **Partner 2: Global Innovation Gathering (GIG)**		
Challenge Category Sustainable urban mobility	**Challenge Title** Citizen-led sensing and data-driven advocacy for safer cycling infrastructure	**Host LAUDS Factory** Université de Lorraine

1. Project Description

The LAUDS Replication Open Call aimed at funding projects by hybrids teams of creatives and technology providers to find solutions to pressing challenges in the mobility, energy, and agriculture/food production sectors. Please provide a description of the project carried out in response to the LAUDS open call that includes the elements listed below.

LAUDS Challenge

DATA-RIDE addressed the mobility challenge by connecting open-source technology, citizen participation, and advocacy. Developed in Nancy in collaboration with Université de Lorraine, the project explored how citizen-generated data could make everyday cycling conditions more visible and support evidence-informed discussions about urban mobility.

The project responded to three interconnected problems identified in its blueprint: everyday cycling experiences are underrepresented in conventional planning data; cycling safety depends on interactions between mobility modes as well as infrastructure; and local advocacy needs accessible evidence to support dialogue with institutions.

Using senseBox:bike devices adapted with a second ToF sensor, volunteers collected measurements during everyday journeys. Spatial analysis and participatory discussions transformed these recordings into shared insights, maps, communication materials, and policy recommendations. The resulting solution combines a technical sensing system with a community-based process for interpreting and communicating evidence.

Project Implementation

[Please describe briefly the project's initially planned objectives and tasks and detail which were successfully achieved, and which were not, as well how and why for both cases. Please also include the planned project timeline, any modifications/adaptations to that were necessary, and their reason.]

The initial proposal described a nine-month programme combining technical development, a three-week sensing campaign, visual analysis, participatory workshops, policy recommendations, and an advocacy toolkit.

The implemented campaign involved ten volunteers collecting data between February and May 2026. The devices recorded GPS position, speed, lateral distance, particulate matter, and motion-based indicators related to stopping and road-surface conditions.

The final report documents 193,328 observations across 357 trips, covering more than 1,000 kilometres and approximately 84 hours of valid recorded movement. Data were cleaned, processed, and linked to OpenStreetMap street segments and selected local datasets shared by the Métropole du Grand Nancy on cycling infrastructure, parking, and tree presence.

Initial objective	Outcome and adaptations
Extend the sensing system with dual-sided distance measurements	The deployed system supported lateral-distance and cycling-corridor analysis. Although the sensors performed successfully during the campaign, further calibration and refinement of the data-processing methods are still needed to improve data accuracy and reliability. Based on challenges identified during the deployment, local partners also supported the development and testing of an alternative mounting prototype for bicycles equipped with rear child seats.
Organise a three-week sensing campaign	Collection took place between February and May, extending a couple of weeks beyond the originally proposed three-week period because some participants were unable to join during the initially scheduled timeframe.
Produce maps and visual indicators	Spatial analyses were developed from sensor recordings and local contextual datasets. Maps were made available by local workshops and presented publicly at the Fête du Vélo 2026.

Co-create policy recommendations	Findings and local experiences were discussed during the Fête du Vélo and a workshop at the Lorraine Fab Living Lab on 1 June. These discussions guided the policy recommendation report.
Toolkit	The project developed communication resources and a blueprint documenting the relationship between sensing, interpretation, advocacy, and sustainability.
Disseminate results openly	Results were presented publicly and shared through project resources. The blueprint is licensed under CC BY 4.0.

The project achieved a locally implemented sensing campaign and a participatory interpretation process. However, the dataset remains exploratory rather than representative of all cycling conditions in Nancy. Overtaking detection also requires further local calibration and video-supported validation. The project produced recommendations, but their adoption by public authorities still has not been demonstrated.

Communication and Dissemination

[Please list and detail the activities and events you have carried out to disseminate your project's activities and results, as well as the LAUDS Factories project as funding source. Please also include details on your strategy to ensure the open reuse of the results stemming from your project.]

Communication formed part of the solution itself, helping participants translate measurements into accessible messages about shared responsibility in public space. Activities and outputs included:

- **Initial workshop, mid-February:** introduction to DATA-RIDE and senseBox:bike, device distribution, installation, and testing.
- **Exploratory workshop, mid-February:** Discussion with local cyclists to identify key topics and hypotheses to guide the data analysis and policy recommendations.
- **Data collection campaign, February–May:** Two rounds of five participants collected data over several weeks using the senseBox:bike equipped with the second ToF sensor.
- **Fête du Vélo:** public presentation and discussion of the project and emerging results.
- **Participatory workshop at Lorraine Fab Living Lab, June :** discussion involving cyclists, researchers, local organisations, and mobility stakeholders.
- **Maps and accompanying presentation:** spatial visualisations connecting findings with streets, corridors, and urban conditions in Nancy.

- **Policy recommendation report:** documentation of measured findings, participant experiences, and proposed directions for action.
- **Communication materials:** Visual materials informed by collective discussions, using the collected data as a basis.
- **Toolkit:** resources explaining the overall process and supporting adaptation by other communities.

The report explicitly identifies DATA-RIDE as developed within the LAUDS Factories programme in collaboration with Université de Lorraine.

Open reuse is supported through documentation of the hardware, software, data workflow, workshop formats, and communication outputs. The blueprint carries a **CC BY 4.0** licence. Other outputs require their respective licences and clear access conditions, particularly for location data.

Team and skill development

[Please list the members of your team that were involved in the project and detail their roles and responsibilities. Indicate what skills and expertise made them suited for their respective role, and mention also skills obtained during OC2.]

Member/entity	Responsibilities and expertise
re:edu	Technical preparation of the senseBox:bike with a second ToF, adaptation of the application and implementation of the French Version - expertise on open hardware, open data platform and software development.
Artur Kim Shum	Coordination, spatial analysis, participatory sensing, and visual communication - expertise on urban planning, and innovation design.
Laura Sobral	Participatory methods, workshop facilitation, collective interpretation, and advocacy - experience in participatory urbanism and collaborative governance.
Global Innovation Gathering (GIG)	Administration, dissemination, and transferability: External communication and knowledge sharing through international open-technology networks, reaching a worldwide community and supporting the transferability of the project.

Université de Lorraine / host team	Local research collaboration, stakeholder engagement, workshop support, laboratory prototyping, volunteer coordination, local communication, and connection with local policymakers. - Anaelle Hily and Laurent Dupont

Collaborative work inside the hybrid team

[Please include a description of the organization of the team involved in the project and the processes that were applied for decision-making throughout the project. Finally, please list and describe any own resources and infrastructures from your team entities that were used during the project. Please indicate how the interaction between the creative and the technology provider parts of the team worked, mention whether and how the LAUDS Factories concept facilitated this collaboration.]

The team worked across technical, participatory, and policy-oriented activities. Technical work focused on developing the sensing system and generating, processing, and analysing the collected data, while the creative and facilitation work helped identify relevant local questions, interpret the findings, and communicate the results.

At the beginning of the project, the hybrid team held regular online meetings to share updates on technical development and coordinate local engagement. The Nancy team played an important bridging role by connecting the project with local stakeholders, cycling organisations, and participants. During the field visits in February and May, the teams jointly facilitated workshops and communicated with participants and local organisations working on cycling and urban mobility.

Throughout the data collection campaign, the Nancy team maintained direct contact with the participants. Challenges encountered during deployment were brought back to the regular team meetings, where possible solutions and adaptations were discussed collectively to allow the campaign to continue.

Decision-making combined different areas of expertise. Technical decisions were primarily led by the re:edu team, while Laura took a stronger role in the policy-related work, always in consultation with the Nancy partners. Participants contributed knowledge about routes, unsafe locations, overtaking conditions, visibility, and infrastructure gaps, while researchers and local actors provided technical and institutional context. All major reports and project outputs were reviewed collectively by the team and can therefore be considered co-authored outputs.

2. Collaboration with LAUDS Factories

The LAUDS Replication Open Call aimed at funding projects by hybrids teams of creatives and technology providers aimed at exploring and working with LAUDS factories. Please provide a description of your collaboration with your host LAUDS factory, as well as with the LAUDS Factories project that includes the elements listed below.

Collaborative Experience with your host

[Please describe your experience in collaborating, interacting, and networking with your host LAUDS Factory, including your experience with your mentor, and your peers within LAUDS Open Call 2. Detail the support you received and role of the LAUDS Factories host in facilitating your project and the development of the proposed solution. Give information on the kind of infrastructure and facilities you have been using in the LAUDS Factory. Indicate what values all this brought to your project.]

Visits and Event

[Please list the dates and duration of your visits to your host LAUDS Factory, as well as any infrastructure, technology, equipment, expertise, and/or skills from the host LAUDS factory you used during your project. Additionally, please include a list of the LAUDS Factories events you participated in and describe in which way they were useful to your project.]

Relation to the researchers and coaches

[Please indicate how your interaction with the LAUDS Factories program outside the host itself (technical and artistic coaches, training and workshop offers, relation to other LAUDS Factories) affected your project.]

Collaborative Experience with your host

Collaboration with Université de Lorraine and the Lorraine Fab Living Lab connected DATA-RIDE with Nancy's research and cycling ecosystem. It supported a process in which sensor evidence could be discussed alongside local experiences and institutional social participation knowledge.

The Lorraine Fab Living Lab hosted the three workshops in February and June, bringing together cyclists, researchers, local organisations, and mobility stakeholders. The place and institution mobilized engaged volunteers and participants for collective interpretation and discussion of possible responses.

Local laboratory support also contributed to development and printing a mounting prototype, addressing a practical aspect of deploying sensors on bicycles with racks.

Visits and Events

Visit/event	Timing	Contribution

Initial workshop and Hypotheses meeting	Mid-February 2026	Introduction to the project, installation and testing of the devices, preparation of volunteers for the data collection campaign, and discussion of the main cycling issues identified by participants.
Fête du Vélo	30 Mai 2026	Public presentation and discussion of results.
Lorraine Fab Living Lab workshop	1 June 2026	Collective interpretation and development of recommendations and communication directions.
Final LAUDS event	October 2026	Exchange and dissemination within the LAUDS network.

Relation to the researchers and coaches

Interactions with the LAUDS coaches took place mainly online during the coaching sessions, with some relevant exchanges focusing on topics closely related to the project, particularly during the mid-term meeting. Towards the end of the project, other LAUDS teams also reached out to exchange experiences and lessons learned regarding fabrication and project implementation.

3. Collaboration with other actors

Explain how your experience with LAUDS Factories helped you to create new collaboration, enter new network or access new communities. Particularly, indicate what local actors you collaborate with as part of the project, including actors you met through the LAUDS Factories host or the events you participated to, or otherwise.

DATA-RIDE connected participating cyclists with researchers, local organisations, and mobility stakeholders through the local team at the University of Lorraine and the Lorraine Smart Cities Living Lab.

These actors contributed at different stages of the project. From the beginning, local organisations showed strong interest in the technology, leading to valuable discussions during the first workshop introducing the senseBox:bike. Several organisations represented in this session also contributed volunteers to the data collection campaign and participated in discussions around hypotheses and key questions concerning cycling conditions in Nancy.

Beyond civil society organisations, representatives from the Métropole du Grand Nancy working on urban mobility participated in the discussions and shared local datasets that could be cross-referenced with the data collected by the project sensors. This collaboration helped connect citizen-generated data with existing information on cycling infrastructure and local urban conditions.

The local network also enabled DATA-RIDE to participate in the Fête du Vélo, a major cycling event in the region, held in a central public square in Nancy. The event provided an opportunity to present the project and its maps to a larger and more diverse audience, discuss the findings with local cyclists and residents, and gather additional feedback on the results.

The local ecosystem included the Métropole du Grand Nancy, Lorraine Smart Cities Living Lab, EDEN, Atelier Dynamo, Laboratoire Sauvage, volunteer cyclists, and workshop participants. Their contributions ranged from providing local knowledge and data to recruiting volunteers, hosting workshops and presentations, supporting the data collection campaign, and facilitating local outreach

Feedback and Recommendations

[Please give us any feedback you see relevant in terms of your experience with the LAUDS Open Call 2, including in terms of your initial expectations versus your lived experience, any challenges you encountered, successes, failures, and lessons learnt.]

Overall, our experience with the LAUDS Open Call 2 organisation was positive, particularly in terms of the support and responsiveness of the LAUDS team. However, there were some challenges regarding the visibility and predictability of the overall process. Deadlines and expected deliverables were not always clear from the beginning, and occasional gaps in communication made it difficult to anticipate when certain outputs would be required. A clearer timeline covering the main milestones, deliverables, and deadlines from the start would have supported project planning.

The online training and capacity-building programme was extensive, but some sessions were less relevant to our project because they addressed topics that did not correspond to its stage of development. Providing a clearer overview of the full programme, including the objectives and expected relevance of each session, could help teams identify when their participation is most valuable.

Despite these challenges, communication with the LAUDS Open Call 2 team was generally constructive. When questions or difficulties emerged, we received useful responses and were

able to find appropriate solutions together. The exchanges also helped us position and communicate the implementation and outcomes of the project more effectively.

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